Boundary limits of the matrix geometric mean and the failure of unrestricted GLT closure

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Abstract

The geometric mean $A\#B$ of two positive definite matrices is the midpoint of the natural geodesic between them and the unique positive solution of the Riccati equation $XA^{-1}X = B$. It is continuous on positive definite pairs but need not be continuous at singular pairs. We describe every limit that can occur: a matrix X is a limit of $A_j\#B_j$ with positive definite $A_j \to A$, $B_j \to B$ exactly when $X \succeq 0$, the block matrix $\left(\begin{smallmatrix} A & X \\ X & B \end{smallmatrix}\right)$ is positive semidefinite, and its rank is at most the ambient dimension. In Ando's description of $A\#B$ as the largest X with that block matrix positive semidefinite, the limits are therefore exactly the positive semidefinite feasible points whose block matrix has rank at most N. A dilation description of the same set, with an auxiliary contraction, already follows from recent work of Jones; our contribution is the explicit elimination of that variable and its consequences. At equal inputs we obtain the full set of limits and the exact interval of limiting Matsumoto fidelities.

For generalized locally Toeplitz (GLT) matrix-sequences we construct uniformly bounded positive definite input sequences with identical scalar symbols whose geometric means have the wrong symbol, or no eigenvalue distribution at all. These examples satisfy the hypotheses of Theorem 4 and Conjecture 1 of the arXiv version of Ilyas, Khan, Loi and Serra-Capizzano and, according to an AI-assisted reading of the published text, of Theorem 3 and Conjecture 1 of their article in Linear Algebra Appl. (2025); these statements are therefore false as stated. We locate the failing step of the published proof. On the positive side, closure holds when the sum of the two symbols is positive definite almost everywhere, for every Kubo–Ando mean (the axiomatic class of operator means that contains the arithmetic, geometric and harmonic means). A new one-sided theorem shows that the regularised symbol is always an upper envelope: the eigenvalues of the means can lose mass relative to it but never gain it. For bounded Riemann-integrable scalar symbols depending on the space variable only, this yields a sharp dichotomy: the symbol formula holds for every positive definite realisation if and only if the two symbols have no common zero set of positive measure or have essentially disjoint supports.

1. Introduction

1.1. The geometric mean and why its boundary matters

For positive numbers a, b the geometric mean $\sqrt{ab}$ is the "multiplicative midpoint" of a and b. Its matrix analogue for Hermitian positive definite (HPD) matrices A, B is

$$A\#B = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2},$$

introduced by Pusz and Woronowicz [1] and placed in the general theory of operator means by Kubo and Ando [2] (a Kubo–Ando mean is any operation on positive matrices satisfying a short list of natural axioms: monotonicity, the transformer inequality and continuity from above; the arithmetic, harmonic and geometric means are examples); see also Ando [3], Lawson and Lim [4] and Bhatia [5]. It keeps the properties one wants from a mean (symmetry, monotonicity, invariance under congruence $A \mapsto TAT^*$) even though A and B need not commute. It is the midpoint of the geodesic joining A and B in the affine-invariant Riemannian geometry of positive definite matrices [5], [6]. It is used to average covariance matrices, for example in brain–computer interfaces [7]; to average structured matrices such as Toeplitz matrices [8]; and in quantum information, where $\operatorname{tr}(\rho\#\sigma)$ is the Matsumoto (or geometric) fidelity of two states [9], [10], [11].

Real data are often rank deficient: covariance matrices with fewer samples than variables, nearly pure quantum states, discretised operators whose coefficients vanish on a region. The mean of singular matrices is defined by a limit, normally $\lim_{\varepsilon\downarrow 0}(A+\varepsilon I)\#(B+\varepsilon I)$, but other approximating sequences can give other limits. That discontinuity is classical. The first question of this paper is: *which* limits can occur? Theorem 1 answers it completely.

1.2. GLT sequences and symbols, intuitively

Discretising a differential or integral operator on finer and finer grids produces a sequence of matrices A_n of growing size. For a large class of such sequences, the *generalized locally Toeplitz* (GLT) class [12], [13], [14], [15], the eigenvalues are asymptotically described by a function $\kappa(x,\theta)$, the *symbol*, of a space variable x and a frequency variable θ: for large n the histogram of the eigenvalues of A_n looks like the histogram of the values of κ. For example, the standard finite-difference matrix of $-u''$ has symbol $2-2\cos\theta$, and a variable coefficient $a(x)$ multiplies it; the diagonal matrix of samples $\operatorname{diag}(a(j/n))_{j=1}^n$ has symbol $a(x)$. GLT sequences form an algebra: sums, products, inverses (when the symbol is invertible almost everywhere) and continuous functions of Hermitian GLT sequences are GLT, and the symbol follows the same operations. Two features matter below. First, the symbol cannot see perturbations whose norm tends to zero, or whose rank is a vanishing fraction of the size. Second, a Hermitian GLT sequence always has a limiting eigenvalue distribution. Two technical terms recur. A sequence is *zero-distributed* if, for every $\delta > 0$, the fraction of its singular values above δ tends to zero; these are exactly the GLT sequences with symbol 0. An *approximating class of sequences* (a.c.s.) is the notion of convergence used in GLT theory: a family of sequences approximates a given one if the differences split into a part of small norm and a part of small relative rank.

The natural question is whether the symbol of $A_n\#B_n$ is the geometric mean of the symbols. If it is, the spectral analysis of an averaged discretisation comes for free.

1.3. What was claimed, and what goes wrong

Ahmad, Khan and Serra-Capizzano [16] proved the symbol formula $\{A_n\#B_n\} \sim_{\mathrm{GLT}} (\kappa\xi)^{1/2}$ for scalar symbols when at least one symbol is nonzero almost everywhere, following Garoni and Serra-Capizzano [12]. Ilyas, Khan, Loi and Serra-Capizzano [17] then stated it with no such hypothesis (we quote the arXiv wording in Section 6.3; an AI-assisted reading of the published version found the same hypotheses), and conjectured the corresponding statement for matrix-valued symbols with the regularised mean as the candidate symbol. The same result appears in Khan's thesis [18].

The unrestricted statement is false. The reason is visible already in dimension two (Section 5): two positive definite matrices can both be within ε of $\operatorname{diag}(1,0)$ while their mean is close to $\operatorname{diag}(c,0)$ for any prescribed $c \in [0,1]$. A coupling of size $O(1/N)$ between the range and

the kernel decides the answer. A GLT symbol cannot see such a coupling. Repeating the two-dimensional example along the diagonal gives HPD sequences with the same step-function symbol whose means have symbol $\frac{4}{5}$ times the predicted one, or no symbol at all (Figure 1, after the table below).

1.4. Results

Result	Statement in brief	Status relative to earlier work
Theorem 1	Limits of $A_j \# B_j$ at a singular pair (A, B) are exactly the $X \succeq 0$ with $\left(\begin{smallmatrix} A & X \\ X & B \end{smallmatrix}\right) \succeq 0$ of rank $\leq N$	Dilation form follows from Jones [19]; explicit elimination new as far as we found
Corollaries 2–3	Equal inputs: limits $0 \preceq X \preceq A$ with $\operatorname{rank}(A - X) \leq \dim \ker A$; exact interval of Matsumoto-fidelity limits	Lower endpoint and the case $q \geq r$ are in Jones; necessity and the full set are new
Section 6	Bounded HPD scalar GLT inputs with equal symbols whose means have the wrong symbol or no distribution	Refutes [17]: arXiv Thm 4, Conj. 1 (quoted); published Thm 3, Conj. 1 (AI-assisted reading, Section 6.3)
Theorem 4	Any sequence of contractions can be hidden behind two fixed input symbols	New as far as we found
Theorem 5, Corollary 6	Closure if $\kappa + \xi \succ 0$ a.e., for every Kubo–Ando mean	Standard identity, GLT transfer
Theorem 7, Corollary 8	Means never gain spectral mass over $\kappa \# \xi$; closure if $\kappa \# \xi = 0$ a.e.	New (sequence-level form of a classical order bound)
Theorem 9, Corollary 10	For bounded Riemann-integrable x-only scalar symbols: closure for all realisations $\iff$ no common zero set of positive measure, or essentially disjoint supports	New
Appendix A	Sharp spectral envelope of means near projections of any rank density	New as far as we found

1.5. Relation to earlier work

Two pieces of earlier work are close and we state the overlap up front. First, Ando [3] and Pusz–Woronowicz [1] characterised $A \# B$ as the *largest* Hermitian X with $\left(\begin{smallmatrix} A & X \\ X & B \end{smallmatrix}\right) \succeq 0$ (Lemma 2.1). Theorem 1 says that the set of boundary limits is exactly the part of this classical feasible set where, in addition, $X \succeq 0$ and the block matrix has rank at most N. In particular every limit lies below $A \# B$.

Second, Jones [19] studies continuity of Kubo–Ando means and channel divergences. His Lemma 4.12.1(b) gives the envelope "every limit point lies below $A \# B$". His Lemma 4.12.3 constructs

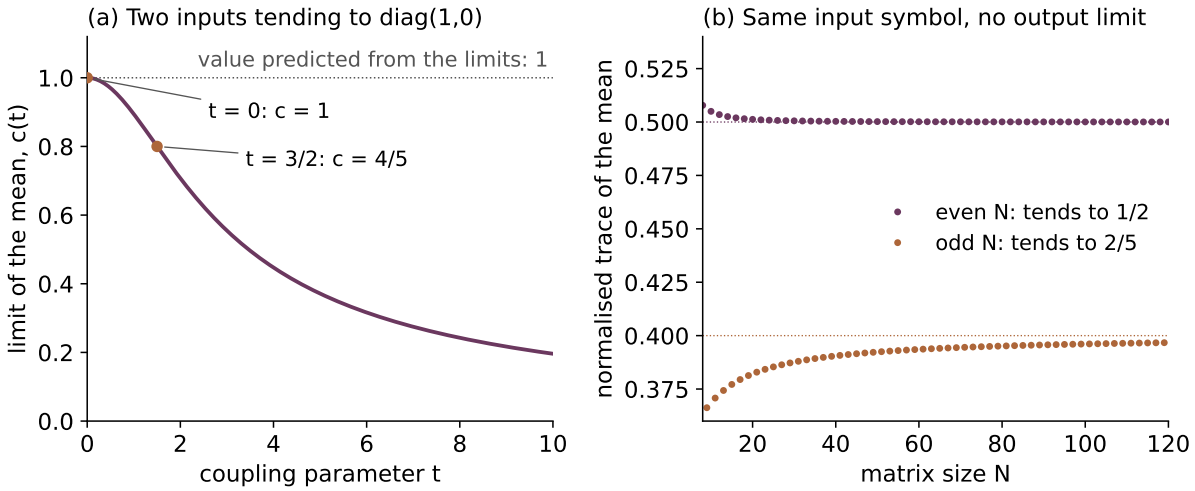


Figure 1. (a) Both inputs $A_e, B_{e,t}$ of (5.1) tend to the projection $\mathrm{diag}(1,0)$ as $e \to 0$, but their geometric mean tends to $c(t)\,\mathrm{diag}(1,0)$ with $c(t) = 2/\sqrt{4+t^2}$: every value in $(0,1]$ occurs. (b) Assembling these blocks gives two matrix-sequences with the same GLT symbol at every size. Choosing $t = 0$ at even sizes and $t = 3/2$ at odd sizes makes the normalised trace of the mean approach $1/2$ along even sizes and $2/5$ along odd sizes, so the means have no limiting eigenvalue distribution. Values are the exact formula (6.6).

boundary limits by dilating a positive contraction into the common kernel; together with an elementary compactness argument this already gives a complete description of the set of limits *with an auxiliary contraction* (Appendix B). His Theorem 4.12.4(b) and Corollary 4.12.5 give the lower endpoint of our fidelity interval and the case in which it is $[0,1]$. The contribution of Theorem 1 is the explicit elimination of the auxiliary contraction into block positivity plus one rank inequality, and the use of that form for the necessity half of the corollaries. The two-dimensional discontinuity itself is classical (an example is attributed to Bhatia [5] in lecture slides by Poulson [20]). The GLT consequences (Sections 6–9) and Appendix A are not in these sources.

1.6. Reader's guide

Section 2 fixes notation and states Theorem 1; Section 3 proves it. Section 4 treats equal inputs and quantum fidelity. Section 5 gives the two-dimensional formula that drives every counterexample. Section 6 contains the GLT counterexamples and locates the error in the published proof. Section 7 shows how much information can be hidden. Section 8 gives the positive closure theorem, and Section 9 the one-sided envelope and the dichotomy, followed by a practical summary ("When is the symbol formula safe?"). Section 10 lists open problems and Section 11 the scope of verification. Appendix A treats growing dimensions at projections; Appendix B records the dilation view and its relation to Theorem 1.

2. Setting and the graph-closure theorem

All matrices are complex and finite dimensional; the real symmetric case uses the same constructions with real data. We write $A \succeq 0$ ($A \succ 0$) for positive semidefinite (definite), $\preceq$ for the Loewner order, and use the operator norm $\|\cdot\|$. The geometric mean of $A, B \succ 0$ is the unique positive definite solution X of

$$XA^{-1}X = B. \tag{2.1}$$

Indeed, conjugation by $A^{-1/2}$ reduces (2.1) to the equation that defines the positive square root of $A^{-1/2}BA^{-1/2}$. Consequently, for every invertible T and all $a, b > 0$,

$$(TAT^*)\#(TBT^*) = T(A\#B)T^*, \qquad (aA)\#(bB) = \sqrt{ab}\,(A\#B). \tag{2.2}$$

The mean is monotone: $A \preceq A'$ and $B \preceq B'$ imply $A\#B \preceq A'\#B'$ [2]. For $A, B \succeq 0$ we use the standard extension

$$A\#B = \lim_{\varepsilon \downarrow 0} (A + \varepsilon I)\#(B + \varepsilon I), \tag{2.3}$$

a decreasing limit by monotonicity. We also use $A\#B \preceq (A + B)/2$: for $A, B \succ 0$ put $T = A^{-1/2}BA^{-1/2}$; then $(T^{1/2} - I)^2 \succeq 0$ gives $T^{1/2} \preceq (I + T)/2$, and congruence by $A^{1/2}$ gives the claim; the singular case follows from (2.3).

For A, B, X Hermitian put

$$M(A, B, X) = \begin{pmatrix} A & X \\ X & B \end{pmatrix}.$$

Lemma 2.1 (classical facts). Let $A, B \succeq 0$.

(a) *(Ando, Pusz–Woronowicz)* $A\#B$ is the largest Hermitian X with $M(A, B, X) \succeq 0$ [3], [1], [5].

(b) $\operatorname{ran}(A\#B) = \operatorname{ran} A \cap \operatorname{ran} B$.

Proof. (a) For $A \succ 0$, $M(A, B, X) \succeq 0$ is equivalent to $B \succeq XA^{-1}X$, that is $Y^2 \preceq K$ with $Y = A^{-1/2}XA^{-1/2}$ and $K = A^{-1/2}BA^{-1/2}$. Since the square root is operator monotone, $Y \preceq |Y| = (Y^2)^{1/2} \preceq K^{1/2}$, which after congruence is $X \preceq A\#B$; equality is feasible by (2.1). For singular A, B, feasibility of X at (A, B) implies feasibility at $(A + \varepsilon I, B + \varepsilon I)$, hence $X \preceq (A + \varepsilon I)\#(B + \varepsilon I)$; let $\varepsilon \to 0$. Feasibility of $A\#B$ itself follows by taking limits. (b) If $u \in \ker A$, the vector $(u, 0)$ has zero quadratic form for the positive semidefinite matrix $M(A, B, A\#B)$, hence lies in its kernel, so $(A\#B)u = 0$; similarly for $\ker B$. Thus $\operatorname{ran}(A\#B) \subseteq \operatorname{ran} A \cap \operatorname{ran} B$. Conversely, if $0 \neq v \in \operatorname{ran} A \cap \operatorname{ran} B$, there is $c > 0$ with $A \succeq c\,vv^*$ and $B \succeq c\,vv^*$, so by monotonicity and (2.2)–(2.3) $A\#B \succeq (c\,vv^*)\#(c\,vv^*) = c\,vv^*$, and $v \in \operatorname{ran}(A\#B)$. □

Theorem 1 (complete graph closure). Let A, B, X be Hermitian $N \times N$ matrices with $A, B \succeq 0$. The following are equivalent.

1. There are $A_j, B_j \succ 0$ with $A_j \to A$, $B_j \to B$ and $A_j\#B_j \to X$.
2. $X \succeq 0$, and

$$M(A, B, X) \succeq 0, \qquad \operatorname{rank} M(A, B, X) \leq N. \tag{2.4}$$

Remarks.

(i) *Each condition is needed.* For $N = 1$, $A = B = 1$ and $X = -1$, the block matrix is positive semidefinite of rank 1 but X is not attainable. For $A = B = I_2$ and $X = \frac{1}{2}I_2$ the block matrix is positive definite (rank $4 > 2$); since $\#$ is continuous at positive definite pairs, the only limit is I_2.

(ii) *Relation to Lemma 2.1(a).* By Theorem 1 every limit is a feasible point for Ando's problem, so every limit satisfies $X \preceq A\#B$ (this envelope is also Jones's Lemma 4.12.1(b) [19]). The regularised mean is the top of the set of limits.

(iii) *Unitary-factor form.* Condition 2 holds if and only if $X \succeq 0$ and $X = A^{1/2}WB^{1/2}$ for some unitary W. Indeed, if (2.4) holds, write $M(A,B,X) = ZZ^*$ with $Z = \left(\begin{smallmatrix} F \\ G \end{smallmatrix}\right)$, $F, G \in \mathbb{C}^{N\times N}$; then $FF^* = A$, $GG^* = B$, $FG^* = X$, and the polar decompositions $F = A^{1/2}U$, $G = B^{1/2}V$ give $X = A^{1/2}UV^*B^{1/2}$. Conversely, $Z = \left(\begin{smallmatrix} A^{1/2} \\ B^{1/2}W^* \end{smallmatrix}\right)$ gives $ZZ^* = M(A,B,X)$, of rank at most N. For $A, B \succ 0$, any such Hermitian X satisfies $XA^{-1}X = B^{1/2}W^*WB^{1/2} = B$, so the set is $\{A\#B\}$, the classical unitary characterisation.

(iv) *Schur-complement reading.* In the normal form of Section 3 the rank condition says that the residual $R = B_0 - CA_0^{-1}C$, a generalised Schur complement in the sense of shorted-operator theory [21], has rank at most the dimension of the common kernel.

Corollary 2 (equal boundary inputs). Let $A \succeq 0$ have rank r in dimension N, and put $q = N - r$. The set of limits of $A_j\#B_j$ over all $A_j, B_j \succ 0$ tending to A is exactly

$$\{X : \ 0 \preceq X \preceq A, \ \operatorname{rank}(A - X) \le q\}. \tag{2.5}$$

In particular, if $q \ge r$, every X with $0 \preceq X \preceq A$ occurs.

3. Proof of Theorem 1

3.1. Necessity

Let $X_j = A_j\#B_j$. By (2.1) the Schur complement of A_j in $M_j = M(A_j, B_j, X_j)$ is $B_j - X_jA_j^{-1}X_j = 0$, so $M_j \succeq 0$ and $\operatorname{rank} M_j = N$. Positive semidefiniteness passes to the limit, and so does rank $\le N$, because all $(N+1)$-minors of M_j vanish. Finally $X \succeq 0$ as a limit of positive definite matrices.

3.2. A simultaneous congruence reduction

Assume condition 2. Put $S = A + B$, $s = \operatorname{rank} S$, $k = N - s$. Choose an invertible T_0 with $T_0ST_0^* = I_s \oplus 0_k$. Since $0 \preceq T_0AT_0^*, T_0BT_0^* \preceq T_0ST_0^*$, both transformed matrices vanish on the last k coordinates; on the first s coordinates they sum to I_s, so they commute and can be diagonalised by one unitary. (No commutation of the original A and B is used.) Grouping eigenvalues equal to 1, equal to 0 and strictly between, we obtain a fixed invertible T with

$$TAT^* = I_a \oplus 0_b \oplus A_0 \oplus 0_k, \quad TBT^* = 0_a \oplus I_b \oplus B_0 \oplus 0_k, \quad A_0, B_0 \succ 0, \ A_0 + B_0 = I_t, \tag{3.1}$$

where $a+b+t+k = N$. By (2.2), both conditions of Theorem 1 are invariant under $(A,B,X) \mapsto (TAT^*, TBT^*, TXT^*)$; block rank is preserved by the block-diagonal congruence $T \oplus T$. So we may assume (3.1).

Support. If $u \in \ker A$, then $(u, 0)$ has zero quadratic form for $M(A,B,X) \succeq 0$, hence lies in its kernel, so $Xu = 0$; similarly for $\ker B$. In (3.1) this forces

$$X = 0_a \oplus 0_b \oplus C \oplus 0_k, \qquad C \succeq 0. \tag{3.2}$$

On the mixed coordinates put $R = B_0 - CA_0^{-1}C$. The Schur complement criterion gives $R \succeq 0$, and block elimination gives

$$\operatorname{rank} M(A,B,X) = a + b + t + \operatorname{rank} R. \tag{3.3}$$

Hence the rank hypothesis is exactly $\operatorname{rank} R \le k$, and we may choose $L \in \mathbb{C}^{t\times k}$ with $R = LL^*$ (padding with zero columns). For real data, T and L can be chosen real.

3.3. The approximating pairs

For $\delta \in (0,1]$ put

$$C_\delta = C + \delta I_t, \qquad Q_\delta = I_k + L^* C_\delta^{-1} L, \tag{3.4}$$

and on the $t+k$ mixed and kernel coordinates

$$\mathcal{A}_\delta = \begin{pmatrix} A_0 & 0 \\ 0 & \delta^4 I_k \end{pmatrix}, \qquad \mathcal{X}_\delta = \begin{pmatrix} C_\delta & \delta^2 L \\ \delta^2 L^* & \delta^4 Q_\delta \end{pmatrix}, \qquad \mathcal{B}_\delta = \mathcal{X}_\delta \mathcal{A}_\delta^{-1} \mathcal{X}_\delta. \tag{3.5}$$

The Schur complement of C_δ in $\mathcal{X}_\delta$ is $\delta^4(Q_\delta - L^* C_\delta^{-1} L) = \delta^4 I_k \succ 0$, so $\mathcal{X}_\delta \succ 0$ and $\mathcal{B}_\delta \succ 0$, and (2.1) gives $\mathcal{A}_\delta \# \mathcal{B}_\delta = \mathcal{X}_\delta$ exactly. Multiplying out,

$$\mathcal{B}_\delta = \begin{pmatrix} C_\delta A_0^{-1} C_\delta + LL^* & \delta^2 (C_\delta A_0^{-1} L + L Q_\delta) \\ \delta^2 (L^* A_0^{-1} C_\delta + Q_\delta L^*) & \delta^4 (L^* A_0^{-1} L + Q_\delta^2) \end{pmatrix}. \tag{3.6}$$

Since $C_\delta \succeq \delta I_t$, $\|Q_\delta\| \le 1 + \|L\|^2/\delta$. Hence $\delta^2 \|L Q_\delta\| \le \delta^2 \|L\| + \delta \|L\|^3$ and $\delta^4 \|Q_\delta\|^2 \le \delta^4 + 2\delta^3 \|L\|^2 + \delta^2 \|L\|^4$. The upper-left block of (3.6) differs from $C A_0^{-1} C + LL^* = B_0$ by $\delta(C A_0^{-1} + A_0^{-1} C) + \delta^2 A_0^{-1}$. Therefore

$$\mathcal{A}_\delta \to A_0 \oplus 0_k, \qquad \mathcal{B}_\delta \to B_0 \oplus 0_k, \qquad \mathcal{X}_\delta \to C \oplus 0_k. \tag{3.7}$$

On each coordinate of the a-block use the scalar pair $(1, \delta^2)$, and on each coordinate of the b-block the pair $(\delta^2, 1)$; both have mean $\delta \to 0$. The direct sum of these pairs with $(\mathcal{A}_\delta, \mathcal{B}_\delta)$, transformed back by T^{-1}, gives the required approximants. Empty blocks are harmless: if $k = 0$ then $R = 0$ and L is void; if $t = 0$ the kernel coordinates carry $\mathcal{A}_\delta = \mathcal{B}_\delta = \mathcal{X}_\delta = \delta^4 I_k$. □

Remark 3.1 (rate). The estimates above show that, in the normal form, the errors are $O(\delta^2)$ for A ($O(\delta^4)$ when $b = 0$; the b-block contributes δ^2) and $O(\delta)$ for B and X, with constants of order $1 + \|C\|\,\|A_0^{-1}\| + \|L\|^3$ (where $\|L\|^2 = \|R\| \le 1$). Undoing the congruence multiplies the constants by at most $\|T^{-1}\|^2$. The constants can be large when A_0 has small eigenvalues; the exact checks in Section 11 confirm the $O(\delta)$ behaviour on complex non-commuting data.

4. Equal inputs and Matsumoto fidelity

Proof of Corollary 2. With $U = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} I & I \\ I & -I \end{smallmatrix} \right)$,

$$U^* M(A, A, X) U = \begin{pmatrix} A + X & 0 \\ 0 & A - X \end{pmatrix}.$$

For $X \succeq 0$, block positivity is therefore $0 \preceq X \preceq A$. Then $\ker(A + X) = \ker A$, so $\operatorname{rank} M(A, A, X) = r + \operatorname{rank}(A - X)$, and Theorem 1 gives (2.5). □

Corollary 3 (limits of the Matsumoto fidelity). Let ρ be a density matrix of rank r in dimension d with nonzero eigenvalues $\lambda_1 \ge \cdots \ge \lambda_r > 0$, and $q = d - r$. Over all positive definite density matrices $\rho_j, \sigma_j \to \rho$, the possible limits of $\operatorname{tr}(\rho_j \# \sigma_j)$ form exactly the interval

$$\Big[\sum_{i=\min(q,r)+1}^{r} \lambda_i,\ 1 \Big], \tag{4.1}$$

where an empty sum is 0. Thus the interval is $\{1\}$ for full rank, and $[0, 1]$ when $q \ge r$.

Proof. By Corollary 2 the matrix limits are $X = \rho - D$ with $0 \preceq D \preceq \rho$ and $\operatorname{rank} D \le q$. If E is the projection onto $\operatorname{ran} D$, then $\operatorname{tr} D \le \operatorname{tr}(E\rho) \le \sum_{i=1}^{\min(q,r)} \lambda_i$ by Ky Fan's maximum principle,

since $\operatorname{rank} E \le q$ and ρ has r nonzero eigenvalues. The bound is attained by the spectral part D_* of ρ for its $\min(q,r)$ largest eigenvalues, and tD_* $(0 \le t \le 1)$ gives every intermediate trace. Theorem 1 supplies positive definite approximants; normalising each by its trace does not change the limit, by (2.2), since both traces tend to 1. Conversely, a convergent sequence of fidelities has a subsequence along which the means converge, because $0 \preceq \rho_j \# \sigma_j \preceq (\rho_j + \sigma_j)/2$ is bounded, and Corollary 2 applies to it. □

The lower endpoint, and the value 0 when $q \ge r$, can be obtained from Jones's dilation results (Theorem 4.12.4(b) with a suitable invariant subspace, and Corollary 4.12.5 [19]), which also prove that the fidelity is continuous at (ρ,ρ) only when ρ has full rank. What Corollaries 2–3 add is necessity: no limit outside (2.5), and no fidelity below (4.1). Cree and Sikora [11] study this fidelity and its semidefinite formulations.

Remark 4.1 (two nearly identical qubit states). Take the matrices A_e and $B_{e,t}$ of (5.1) and normalise them: $\rho_e = A_e/(1+e^2)$ and $\sigma_e = B_{e,t}/(1+(1+t^2)e^2)$. Here ρ_e is the pure state $|0\rangle$ with noise of weight e^2 in the other direction, and $B_{e,t} = vv^* + \operatorname{diag}(0,e^2)$ with $v = (1,te)$ is a pure state rotated by an angle of order te plus the same noise. Both states tend to $|0\rangle\langle 0|$ as $e \to 0$, so their trace distance tends to 0 and their Uhlmann fidelity $\left(\operatorname{tr}|\rho_e^{1/2}\sigma_e^{1/2}|\right)^2$ [22] tends to 1 (it is continuous and equals 1 at the limit). By (5.2), however,

$$\operatorname{tr}(\rho_e \# \sigma_e) = \frac{2+(2+t^2)e^2}{\sqrt{4+t^2}\,\sqrt{(1+e^2)(1+(1+t^2)e^2)}} \longrightarrow \frac{2}{\sqrt{4+t^2}}.$$

So two states that are arbitrarily close can have any Matsumoto fidelity in $(0,1]$. The trigger is not exotic: a coherent rotation of size $\theta = te$ combined with a noise floor of order θ^2. Every state in this family has full rank, so positive definiteness alone is no safeguard. Matsumoto fidelity should therefore not be used as a robust closeness measure near pure states without quantitative control of the smallest eigenvalues, for example a uniform positive lower bound on the spectrum.

5. A two-dimensional formula

For $e > 0$ and $t \ge 0$ let

$$A_e = \begin{pmatrix} 1 & 0 \\ 0 & e^2 \end{pmatrix}, \qquad B_{e,t} = \begin{pmatrix} 1 & te \\ te & (1+t^2)e^2 \end{pmatrix}. \tag{5.1}$$

Both are positive definite with determinant e^2. A direct multiplication gives $(A_e + B_{e,t})A_e^{-1}(A_e + B_{e,t}) = (4+t^2)B_{e,t}$, so $X = (A_e + B_{e,t})/\sqrt{4+t^2}$ is positive definite and satisfies $XA_e^{-1}X = B_{e,t}$. By (2.1),

$$A_e \# B_{e,t} = \frac{1}{\sqrt{4+t^2}} \begin{pmatrix} 2 & te \\ te & (2+t^2)e^2 \end{pmatrix}. \tag{5.2}$$

For fixed t, both inputs tend to $P = \operatorname{diag}(1,0)$, while

$$A_e \# B_{e,t} \to c(t)P, \qquad c(t) = \frac{2}{\sqrt{4+t^2}}. \tag{5.3}$$

The value 0 is also attainable, letting $t \to \infty$ with $te \to 0$. At $t = 3/2$ every coefficient is rational:

$$B_{e,3/2} = \begin{pmatrix} 1 & 3e/2 \\ 3e/2 & 13e^2/4 \end{pmatrix}, \qquad A_e \# B_{e,3/2} = \begin{pmatrix} 4/5 & 3e/5 \\ 3e/5 & 17e^2/10 \end{pmatrix}. \tag{5.4}$$

Formula (5.2) is a transparent instance of the classical mechanism; no novelty is claimed for it.

6. Scalar GLT counterexamples

6.1. The GLT facts used

We use the following results for (block, multilevel) GLT sequences, with the numbering of Barbarino, Garoni and Serra-Capizzano [14], [15]; the scalar unilevel versions are in Garoni and Serra-Capizzano [12], [13]. Approximating classes of sequences (a.c.s.) go back to [23]; see also [24].

- (G1) A Hermitian GLT sequence has its symbol as eigenvalue distribution (axiom GLT 1).
- (G2) Diagonal sampling of an almost-everywhere continuous bounded function a (equivalently, a Riemann-integrable one) has symbol a: [14], Theorem 4.24, p. 78 (unilevel) and [15], Theorem 4.23, p. 169 (multilevel); also axiom GLT 3, second item.
- (G3) A sequence is zero-distributed if and only if it is GLT with symbol 0 (axiom GLT 3, third item: [14], p. 86; [15], p. 180). In particular adding a sequence whose norm tends to zero does not change a symbol.
- (G4) Sums and products: [14], Theorem 4.17, p. 73; [15], Theorem 4.16, p. 164.
- (G5) Pseudoinverse: [14], Theorem 4.18, p. 73, which reads "If $\{A_n\}_n \sim_{\mathrm{GLT}} \kappa$ and κ is invertible a.e., then $\{A_n^\dagger\}_n \sim_{\mathrm{GLT}} \kappa^{-1}$"; multilevel: [15], Theorem 4.17, p. 165.
- (G6) Continuous functional calculus: [14], Theorem 4.19, p. 74; [15], Theorem 4.18, p. 165, which reads "If $\{A_n\}_n \sim_{\mathrm{GLT}} \kappa$ and each A_n is Hermitian, then $\{f(A_n)\}_n \sim_{\mathrm{GLT}} f(\kappa)$ for any continuous function $f : \mathbb{C} \to \mathbb{C}$." No uniform norm bound on A_n is required.

6.2. The construction

Let $a = \mathbf{1}_{(0,1/2]}$ on $[0,1]$ and, for each size N, $m = \lfloor N/2 \rfloor$, so that the diagonal sampling is $P_N = D_N(a) = \mathrm{diag}(a(j/N))_{j=1}^N = \mathrm{diag}(I_m, 0_{N-m})$. Take $e_N = 1/N$, $d_N = N - 2m \in \{0,1\}$, a bounded sequence $t_N \geq 0$, and

$$A_N = \begin{pmatrix} I_m & 0 \\ 0 & e_N^2 I_m \end{pmatrix} \oplus e_N^2 I_{d_N}, \qquad B_N = \begin{pmatrix} I_m & t_N e_N I_m \\ t_N e_N I_m & (1+t_N^2) e_N^2 I_m \end{pmatrix} \oplus e_N^2 I_{d_N}. \tag{6.1}$$

Both are positive definite (their Schur complements are $e_N^2 I_m$), uniformly bounded, and

$$\|A_N - P_N\| = e_N^2, \qquad \|B_N - P_N\| \leq t_N e_N + (1+t_N^2) e_N^2 \to 0. \tag{6.2}$$

By (G2) and (G3), $\{A_N\} \sim_{\mathrm{GLT}} a$ and $\{B_N\} \sim_{\mathrm{GLT}} a$. After a permutation the matrices are direct sums of copies of (5.1), so (5.2) gives

$$A_N \# B_N = \frac{1}{\sqrt{4+t_N^2}} \begin{pmatrix} 2I_m & t_N e_N I_m \\ t_N e_N I_m & (2+t_N^2) e_N^2 I_m \end{pmatrix} \oplus e_N^2 I_{d_N}, \tag{6.3}$$

and in particular

$$\|A_N \# B_N - c(t_N) P_N\| \leq \frac{t_N e_N + (2+t_N^2) e_N^2}{\sqrt{4+t_N^2}} + e_N^2 \to 0. \tag{6.4}$$

A wrong output symbol. With $t_N = 3/2$, (6.4), (G2) and (G3) give

$$\{A_N\} \sim_{\mathrm{GLT}} a, \qquad \{B_N\} \sim_{\mathrm{GLT}} a, \qquad \{A_N \# B_N\} \sim_{\mathrm{GLT}} \tfrac{4}{5} a, \tag{6.5}$$

whereas $(a \cdot a)^{1/2} = a$, and the regularised mean of the symbols (the candidate in Conjecture 1 of [17]) is also a.

No output distribution. With $t_N = 0$ for even N and $t_N = 3/2$ for odd N, the normalised trace is exactly

$$\frac{1}{N}\operatorname{tr}(A_N \# B_N) = \frac{m}{N} \cdot \frac{2 + (2 + t_N^2)e_N^2}{\sqrt{4 + t_N^2}} + \frac{d_N}{N} e_N^2, \tag{6.6}$$

which tends to $1/2$ along even N and to $2/5$ along odd N. The empirical eigenvalue distributions tend to $\frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and $\frac{1}{2}\delta_0 + \frac{1}{2}\delta_{4/5}$ respectively. Since the means are uniformly bounded, a continuous compactly supported test function equal to x on a common spectral interval distinguishes the two limits, so $\{A_N \# B_N\}$ has no eigenvalue distribution and, by (G1), is not GLT at all. The sequence is defined at every size N.

6.3. The statements refuted, and where the proof fails

arXiv version. We read arXiv:2505.03256v1 of [17] in full. Its Theorem 4 (printed p. 11) reads, in part: "Let $r = 1$ and $d \geq 1$. Assume $\{A_n\}_n \sim_{\mathrm{GLT}} \kappa$ and $\{B_n\}_n \sim_{\mathrm{GLT}} \xi$, where $A_n, B_n \in \mathcal{P}_{\nu(n)}$ for every multi-index n. Then $\{G(A_n, B_n)\}_n \sim_{\mathrm{GLT}} (\kappa\xi)^{1/2}$", where $\mathcal{P}_{\nu(n)}$ denotes the HPD matrices of that size and G the geometric mean; a spectral-distribution relation is also asserted. No boundedness, structure or regularisation hypothesis is made. The sequences (6.1) satisfy these hypotheses with $d = 1$ and $\kappa = \xi = a$, so (6.5) contradicts Theorem 4. Conjecture 1 (printed p. 21) begins: "Let $\{A_n\}_n$ and $\{B_n\}_n$ be Hermitian positive definite GLT matrix-sequences with $\{A_n\}_n \sim_{\mathrm{GLT}} \kappa$, $\{B_n\}_n \sim_{\mathrm{GLT}} \xi$. Then the geometric mean matrix-sequence is again GLT", with the candidate symbol of Definition 3, equation (4.5) (p. 21), the regularised mean of the symbols. For $\kappa = \xi = a$ the candidate is a, so (6.5) and the alternating example contradict it. *Warning on numbering:* the arXiv version's "Theorem 3" is a restatement of the earlier, correct result of Ahmad et al. [16] (which assumes that one symbol is nonzero almost everywhere); it is not affected.

Published version (Linear Algebra Appl., 2025, doi:10.1016/j.laa.2025.08.004). The scalar theorem is Theorem 3 (statement from p. 8, proof from p. 9); Conjecture 1 is on p. 19, with the extended mean in Definition 3, equation (4.5), p. 18. Our own source record transcribed only short passages verbatim. A separate AI-assisted reading of the published institutional full text confirmed that the hypotheses of published Theorem 3 (pp. 8–9) and of Conjecture 1 (p. 19) admit the examples above, and that the published proof contains the identification attacked below (p. 10). We record this as an AI-assisted reading, not a transcription; readers can check the cited pages. In both versions, after showing that the shifted means $\{G(A_n + \varepsilon I, B_n)\}$ form an a.c.s. for a limit sequence, the proof asserts (published p. 10; arXiv p. 12, where the text has "in the a.c.s. topology")

> "and this limit coincides with $\{G(A_n, B_n)\}_n$ in a.c.s. topology".

The examples show that this identification is false. Concretely, in the arXiv proof the limit sequence is $\{(A_n B_n^2 A_n)^{1/4}\}$, and any argument identifying it with $\{G(A_n, B_n)\}$ up to a zero-distributed sequence must fail here: in (6.1) with $t_N = 3/2$ we have $(A_N B_N^2 A_N)^{1/4} \to P_N$ in norm (the blocks converge uniformly and $x \mapsto x^{1/4}$ is continuous on bounded positive matrices), while $A_N \# B_N \to \frac{4}{5} P_N$; the difference $\frac{1}{5} P_N$ has rank $m \sim N/2$ and is not zero-distributed. The same problem affects the heuristic given for Conjecture 1, which shifts both inputs and passes to the limit $\varepsilon \to 0$ after $n \to \infty$.

Related statements. Immediately before its Theorem 3 the published article states that the result proves a Conjecture 10.1, in the words "thus proving [21], Conjecture 10.1." (p. 8; the

"[21]" is that article's own reference number); Khan's thesis [18] attributes this conjecture to the Garoni–Serra-Capizzano monograph [12] immediately before its Theorem 5.2 (printed p. 66). We could not access the relevant pages of the monograph and do not quote or claim to refute its Conjecture 10.1 as worded there. If, as these later sources indicate, it asserts the scalar symbol formula without a non-vanishing hypothesis, then the examples above refute it. The thesis presents its Theorem 5.2 (printed p. 66), whose conclusion reads "$\{G(A_n, B_n)\}_n \sim_{\mathrm{GLT}} (\kappa\xi)^{1/2}$", as the same unrestricted scalar result, and its Conjecture 5.2.1 (printed p. 77, candidate mean in Definition 3, equation (5.13), p. 76) states "Then the geometric mean matrix-sequence is again GLT and" the candidate formula; both are affected in the same way. The restricted theorem of Ahmad et al. ([16], Theorem 3, p. 10, attributed there to Theorem 10.2 of [12]), with the hypothesis "Assume that at least one between κ and ξ is nonzero almost everywhere.", is not contradicted: in (6.1) both symbols vanish on $(1/2, 1]$.

6.4. How structured are the counterexamples?

Remark 6.1 (the perturbation is small but non-local). The perturbation of P_N in (6.1) has norm $O(1/N)$, so it is invisible to the symbol; but the off-diagonal block couples the sample point x with $x + 1/2$. It is therefore not of the form $g(n)T_n(f)$ with a fixed symbol f and $g(n) \to 0$, which is the kind of small-norm term that arises in discretisations. The refutation is of the theorems as stated, which allow any HPD GLT sequences.

Remark 6.2 (a fully structured block counterexample). With 2×2 symbols ($r = 2$) no permutation is needed. Let

$$\mathcal{A}_n = I_n \otimes A_{e_n}, \qquad \mathcal{B}_n = I_n \otimes B_{e_n,t}, \qquad e_n \to 0,\ t = 3/2.$$

For a constant matrix F the block Toeplitz matrix $T_n(F)$ is $I_n \otimes F$, so $\mathcal{A}_n = T_n(F_0) + e_n^2 T_n(F_2)$ and $\mathcal{B}_n = T_n(F_0) + e_n t\, T_n(F_1) + e_n^2(1+t^2)T_n(F_2)$ with $F_0 = \mathrm{diag}(1, 0)$, $F_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $F_2 = \mathrm{diag}(0, 1)$. Both sequences are HPD with the constant symbol F_0, and the perturbations have exactly the traditional form $g(n)T_n(f)$. By (5.2), $\mathcal{A}_n \# \mathcal{B}_n = I_n \otimes (A_{e_n} \# B_{e_n,t})$ has symbol $\frac{4}{5}F_0$, whereas the candidate symbol of Conjecture 1 (and $(\kappa\xi)^{1/2}$ for these commuting symbols) is F_0. This refutes the block form of Conjecture 1 natively, and also the block theorem for commuting symbols in the arXiv version (Theorem 6, p. 13), which we checked only in that version.

Whether closure holds for scalar sequences generated by fixed-symbol Toeplitz matrices, diagonal samplings and terms $g(n)T_n(f)$ is open (Section 10).

7. Arbitrary contraction data behind fixed symbols

Theorem 4 (universal contraction realisation). For every $N \geq 2$ let $m = \lfloor N/2 \rfloor$ and let $C^{(N)}$ be any Hermitian $m \times m$ matrix with $0 \preceq C^{(N)} \preceq I_m$. There are positive definite A_N, B_N with

$$\|A_N - P_N\| + \|B_N - P_N\| \to 0, \qquad \left\| A_N \# B_N - \left(C^{(N)} \oplus 0_{N-m}\right) \right\| \to 0, \tag{7.1}$$

where $P_N = D_N(a)$ as in Section 6. In particular both input sequences are GLT with symbol a, while the output can encode any contraction sequence, including one with no spectral limit.

Proof. First let $N = 2n$. Choose $0 < \delta_n \leq 1$ with $\delta_n \to 0$, and put

$$D_n = \frac{C^{(N)} + \delta_n I_n}{1 + \delta_n}, \qquad e_n = \delta_n^2, \qquad T_n = 2\big(D_n^{-2} - I_n\big)^{1/2}. \tag{7.2}$$

Then $0 \prec D_n \preceq I_n$, $\|D_n - C^{(N)}\| \le \delta_n$ and $\|T_n\| \le 2(1+\delta_n)/\delta_n$. Set

$$A_N = \begin{pmatrix} I_n & 0 \\ 0 & e_n^2 I_n \end{pmatrix}, \qquad B_N = \begin{pmatrix} I_n & e_n T_n \\ e_n T_n & e_n^2 (I_n + T_n^2) \end{pmatrix}. \tag{7.3}$$

The Schur complement of the upper-left block of B_N is $e_n^2 I_n$, so $B_N \succ 0$, and $\|B_N - P_N\| \le e_n\|T_n\| + e_n^2(1 + \|T_n\|^2) \le 4\delta_n + 16\delta_n^2 + \delta_n^4 \to 0$. All blocks are functions of D_n; diagonalising D_n reduces the mean to (5.2) with $t =$ the eigenvalues of T_n. Since $\sqrt{4I + T_n^2} = 2D_n^{-1}$ and $\frac{1}{2} D_n T_n = (I - D_n^2)^{1/2}$,

$$A_N \# B_N = \begin{pmatrix} D_n & e_n (I_n - D_n^2)^{1/2} \\ e_n (I_n - D_n^2)^{1/2} & e_n^2 (2D_n^{-1} - D_n) \end{pmatrix}. \tag{7.4}$$

The off-diagonal blocks have norm at most e_n, and the lower-right block at most $e_n^2(2(1+\delta_n)/\delta_n + 1) \le 5\delta_n^3$. With $\|D_n - C^{(N)}\| \le \delta_n$ this gives (7.1). For $N = 2n+1$ append one coordinate on which A_N, B_N and their mean all equal e_n^2; then $P_N = \operatorname{diag}(I_n, 0_{n+1}) = D_N(a)$ still, and (7.1) follows from the even case. □

If $C^{(N)}$ has a limiting spectral law ν, the means have law $\frac{1}{2}\nu + \frac{1}{2}\delta_0$. Appendix A shows that this is the most freedom possible at a projection of rank density $1/2$.

8. Closure when the sum symbol is positive definite

Theorem 5. Let $\{A_n\}$, $\{B_n\}$ be HPD (scalar or $r \times r$ block, unilevel or multilevel) GLT sequences of the same sizes with symbols κ, ξ (Hermitian positive semidefinite a.e.). If

$$\Sigma := \kappa + \xi \succ 0 \quad \text{almost everywhere,} \tag{8.1}$$

then

$$\{A_n \# B_n\} \sim_{\mathrm{GLT}} \Sigma^{1/2} \big[H(I-H)\big]^{1/2} \Sigma^{1/2}, \qquad H = \Sigma^{-1/2} \kappa \Sigma^{-1/2}, \tag{8.2}$$

and the right-hand side equals $\kappa \# \xi$ (defined by (2.3)) almost everywhere; in the scalar case it is $\sqrt{\kappa\xi}$.

Proof. Put $S_n = A_n + B_n \succ 0$ and $H_n = S_n^{-1/2} A_n S_n^{-1/2}$, so that $0 \prec H_n \prec I$. By (2.2) with $T = S_n^{1/2}$,

$$A_n \# B_n = S_n^{1/2} (H_n \# (I - H_n)) S_n^{1/2} = S_n^{1/2} \big[H_n (I - H_n)\big]^{1/2} S_n^{1/2}, \tag{8.3}$$

because H_n and $I - H_n$ commute. By (G4), $\{S_n\} \sim_{\mathrm{GLT}} \Sigma$. Since Σ is invertible a.e. and S_n is invertible, (G5) gives $\{S_n^{-1}\} = \{S_n^{\dagger}\} \sim_{\mathrm{GLT}} \Sigma^{-1}$. Applying (G6) with $f(x) = \sqrt{\max(\operatorname{Re} x, 0)}$ to the Hermitian sequences S_n and S_n^{-1} gives $\{S_n^{1/2}\} \sim_{\mathrm{GLT}} \Sigma^{1/2}$ and $\{S_n^{-1/2}\} = \{(S_n^{-1})^{1/2}\} \sim_{\mathrm{GLT}} \Sigma^{-1/2}$; no norm bound is needed, although S_n^{-1} may be unbounded. By (G4), $\{H_n\} \sim_{\mathrm{GLT}} H$ and $\{H_n(I - H_n)\} \sim_{\mathrm{GLT}} H(I-H)$; the matrices $H_n(I-H_n)$ are Hermitian because the factors commute. One more application of (G6) and (G4) gives (8.2).

For the identification, fix a point where $\Sigma \succ 0$ and apply (8.3) to $\kappa + \varepsilon I$, $\xi + \varepsilon I$: their sum $\Sigma + 2\varepsilon I$ tends to $\Sigma \succ 0$, so every factor converges (the inverse square root at the invertible limit, and the square root on the closed positive cone), and the limit is the right side of (8.2). By (2.3) the left side tends to $\kappa \# \xi$. □

The sum normalisation (8.3) is standard (it is the coordinate used by Jones [19], §4.12); the content of Theorem 5 is its GLT transfer. Hypothesis (8.1) is weaker than invertibility of one entire symbol: complementary indicator symbols satisfy it although neither is invertible a.e.

Corollary 6 (every Kubo–Ando mean). Under the hypotheses of Theorem 5, let σ be a Kubo–Ando mean with representing function f (positive and operator monotone on $(0,\infty)$), so that $A\sigma B = A^{1/2} f(A^{-1/2} B A^{-1/2}) A^{1/2}$ for $A, B \succ 0$. Define

$$\phi(t) = t\, f\Big(\frac{1-t}{t}\Big)\ (0<t<1), \qquad \phi(0) = \lim_{u\to\infty} \frac{f(u)}{u}, \qquad \phi(1) = f(0+), \tag{8.4}$$

and extend ϕ to $\mathbb{R}$ by the constant $\phi(0)$ on $(-\infty, 0]$ and $\phi(1)$ on $[1,\infty)$, and to $\mathbb{C}$ by $z \mapsto \phi(\operatorname{Re} z)$. Then $\{A_n \sigma B_n\} \sim_{\mathrm{GLT}} \Sigma^{1/2} \phi(H) \Sigma^{1/2} = \kappa \sigma \xi$.

Proof. A positive operator monotone function is concave and nondecreasing, so $f(0+)$ exists and $f(u)/u$ is nonincreasing and nonnegative, so its limit at ∞ exists; with $u = (1-t)/t$, $\phi(t) = f(u)/(1+u)$, so ϕ is continuous on $[0,1]$, and its extension is continuous on $\mathbb{C}$, as (G6) requires. (The normalisation $f(1) = 1$ is not needed.) Congruence invariance gives $A_n \sigma B_n = S_n^{1/2} \phi(H_n) S_n^{1/2}$, and the proof of Theorem 5 applies with (G6) for ϕ; the spectra of H_n lie in $(0,1)$, so the extension does not change $\phi(H_n)$. The identification with the regularised mean is as before. □

9. One-sided domination and a sharp dichotomy

The common shift in (2.3) produces an *upper* bound by monotonicity. At the level of matrix-sequences this gives a one-sided theorem that holds with no hypothesis on the symbols.

Notation: for a d-level, $r \times r$ block sequence of size d_n, the symbols are defined on $D = [0,1]^d \times [-\pi,\pi]^d$. For a measurable Hermitian-positive-semidefinite-valued g on D let ν_g be the probability measure on $[0,\infty)$ given by $\nu_g(E) = \frac{1}{(2\pi)^d r} \int_D \#\{i : \lambda_i(g(x,\theta)) \in E\}\, dx\, d\theta$, and write μ_n for the empirical eigenvalue measure of an n-th matrix.

Theorem 7 (the regularised symbol is a universal upper envelope). Let $\{A_n\} \sim_{\mathrm{GLT}} \kappa$ and $\{B_n\} \sim_{\mathrm{GLT}} \xi$ be HPD (scalar or block, uni- or multilevel) and $G_n = A_n \# B_n$. Let $\gamma = \kappa \# \xi$ pointwise, as in (2.3). Then:

(a) for every $\delta > 0$,
$$\limsup_{n\to\infty} \frac{1}{d_n} \#\{j : \lambda_j(G_n) \ge \delta\} \le \nu_\gamma([\delta,\infty)); \tag{9.1}$$

(b) the measures μ_n of G_n are tight, and every subsequential weak limit ν is stochastically dominated by ν_γ: $\nu((\delta,\infty)) \le \nu_\gamma((\delta,\infty))$ for every $\delta \in \mathbb{R}$.

In words: the eigenvalues of the means can lose mass relative to the regularised symbol, but never gain it.

Proof. (a) Fix $\varepsilon > 0$. The sequence $\{\varepsilon I\}$ is GLT with symbol εI_r, so by (G4) $\{A_n + \varepsilon I\} \sim_{\mathrm{GLT}} \kappa + \varepsilon I$ and similarly for B_n. The sum symbol is $\succeq 2\varepsilon I \succ 0$, so Theorem 5 gives $\{G_n^\varepsilon\} \sim_{\mathrm{GLT}} \gamma_\varepsilon$, where $G_n^\varepsilon = (A_n + \varepsilon I) \# (B_n + \varepsilon I)$ and $\gamma_\varepsilon = (\kappa + \varepsilon I) \# (\xi + \varepsilon I)$. By (G1) the empirical measures μ_n^ε of G_n^ε converge vaguely to the probability measure ν_{γ_ε} (γ_ε is finite a.e.); vague convergence of probability measures to a probability measure is weak convergence. Monotonicity of $\#$ gives $G_n \preceq G_n^\varepsilon$, so by Weyl's monotonicity principle the j-th largest eigenvalues satisfy $\lambda_j^{\downarrow}(G_n) \le \lambda_j^{\downarrow}(G_n^\varepsilon)$ and hence $\#\{j : \lambda_j(G_n) \ge \delta\} \le \#\{j : \lambda_j(G_n^\varepsilon) \ge \delta\}$. By the portmanteau theorem for the closed set $[\delta,\infty)$,

$$\limsup_n \frac{1}{d_n} \#\{j : \lambda_j(G_n) \ge \delta\} \le \limsup_n \mu_n^\varepsilon([\delta,\infty)) \le \nu_{\gamma_\varepsilon}([\delta,\infty)).$$

Now let $\varepsilon \downarrow 0$. At every point, γ_ε decreases to γ in the Loewner order, so the ordered eigenvalues satisfy $\lambda_i(\gamma_\varepsilon) \ge \lambda_i(\gamma)$ and, by continuity of ordered eigenvalues, $\lambda_i(\gamma_\varepsilon) \to \lambda_i(\gamma)$. Hence $\mathbf{1}\{\lambda_i(\gamma_\varepsilon) \ge$

$\delta\} \to \mathbf{1}\{\lambda_i(\gamma) \geq \delta\}$ pointwise: if $\lambda_i(\gamma) \geq \delta$ both indicators equal 1 for every ε, and if $\lambda_i(\gamma) < \delta$ the first is eventually 0. No assumption on where κ or ξ vanishes is needed. Dominated convergence gives $\nu_{\gamma_\varepsilon}([\delta, \infty)) \to \nu_\gamma([\delta, \infty))$, which proves (9.1).

(b) Since $G_n \succ 0$ and (9.1) with $\delta = M$ tends to 0 as $M \to \infty$ (γ is finite a.e.), the μ_n are tight. Let ν be a subsequential weak limit and $\delta' > 0$. The portmanteau theorem for the open set (δ', ∞) and (9.1) give $\nu((\delta', \infty)) \leq \liminf \mu_n((\delta', \infty)) \leq \nu_\gamma([\delta', \infty))$. For $\delta \geq 0$ and $\delta' \downarrow \delta$, the left side increases to $\nu((\delta, \infty))$ and the right side increases to $\nu_\gamma((\delta, \infty))$, since the sets $[\delta', \infty)$ increase to (δ, ∞). For $\delta < 0$ both sides equal 1. □

The fixed-dimensional analogue (every limit point lies below $A\#B$) is Lemma 2.1(a) and Jones's Lemma 4.12.1(b) [19], whose proof uses the same shift. The published article of Ilyas et al. calls one of its own results maximal, in the sense of its conclusion (p. 27): "This shows that Theorem 5 is maximal" (their Theorem 5, pp. 10–11, whose statement we did not transcribe); our source record located no one-sided spectral comparison between $G(A_n, B_n)$ and $(\kappa\xi)^{1/2}$ there, so we present Theorem 7 as new, within the bounds of that check.

Corollary 8 (disjoint supports give closure). If $\kappa\#\xi = 0$ almost everywhere, then $\{A_n\#B_n\}$ is zero-distributed, hence $\{A_n\#B_n\} \sim_{\mathrm{GLT}} 0 = \kappa\#\xi$. By Lemma 2.1(b) the hypothesis means $\operatorname{ran}\kappa \cap \operatorname{ran}\xi = \{0\}$ a.e.; for scalar symbols it means $\kappa\xi = 0$ a.e.

Proof. Here $\nu_\gamma = \delta_0$, so (9.1) gives $d_n^{-1}\#\{j : \lambda_j(G_n) \geq \delta\} \to 0$ for every $\delta > 0$. As $G_n \succ 0$, its singular values are its eigenvalues, so $\{G_n\}$ is zero-distributed, and (G3) applies. □

Together, Theorem 5 and Corollary 8 cover two situations; the next theorem shows that, for scalar symbols of the space variable, nothing else is safe.

Theorem 9 (failure whenever a common zero set meets a common support). Let $d \geq 1$ and let $\kappa, \xi : [0,1]^d \to [0, \infty)$ be bounded and Riemann integrable, regarded as GLT symbols independent of θ. Let $|\cdot|$ denote Lebesgue measure. If

$$|\{\kappa + \xi = 0\}| > 0 \qquad \text{and} \qquad |\{\kappa\xi > 0\}| > 0, \tag{9.2}$$

then there are HPD sequences $\{A_n\} \sim_{\mathrm{GLT}} \kappa$ and $\{B_n\} \sim_{\mathrm{GLT}} \xi$ with $\sup_n(\|A_n\| + \|B_n\|) < \infty$ such that

(i) $\limsup_n d_n^{-1} \operatorname{tr}(A_n\#B_n) < \int_{[0,1]^d} \sqrt{\kappa\xi}\, dx$, so $\{A_n\#B_n\}$ is not GLT with symbol $\sqrt{\kappa\xi}$; and

(ii) after a modification along every other index, $\{A_n\#B_n\}$ has no eigenvalue distribution at all.

Proof. We write the proof for $d = 1$, sizes n and sample points $x_j = j/n$. For $d > 1$ the sequence is indexed by multi-indices $n = (n_1, \ldots, n_d)$ with $\min_i n_i \to \infty$, the sample points are $x_j = j/n$ (componentwise) for j in the grid $\{1, \ldots, n_1\} \times \cdots \times \{1, \ldots, n_d\}$, and the size is $N(n) = n_1 \cdots n_d$. The proof uses only two facts about the grid, and both hold for $d > 1$: Riemann sums of a Riemann-integrable function on $[0,1]^d$ over this grid converge to its integral, because the mesh $\max_i 1/n_i$ tends to zero; and diagonal sampling of a Riemann-integrable function has that function as symbol (the multilevel form of (G2), [15], Theorem 4.23). With n replaced by $N(n)$ and index sets taken in the grid, every step below holds verbatim. Let $K \geq \sup(\kappa + \xi)$, $K \geq 1$, and write $\kappa_j = \kappa(x_j)$, $\xi_j = \xi(x_j)$, $s_j = \sqrt{\kappa_j\xi_j}$.

Step 1: counting sample indices. For continuous bounded $F : \mathbb{R}^2 \to \mathbb{R}$ the function $F(\kappa, \xi)$ is bounded and continuous wherever both κ and ξ are, hence Riemann integrable, so $n^{-1}\sum_j F(\kappa_j, \xi_j) \to \int_0^1 F(\kappa, \xi)$. Thus the empirical law of the pairs (κ_j, ξ_j) converges weakly

to the law π of $(\kappa(x), \xi(x))$ under Lebesgue measure, and the portmanteau theorem gives $\liminf_n \pi_n(U) \geq \pi(U)$ for open $U \subset \mathbb{R}^2$. (No continuity-point condition is needed for open sets.) Put $z = |\{\kappa + \xi = 0\}| > 0$ and choose $\eta_0 > 0$ with $w = |\{\kappa\xi > \eta_0\}| > 0$, which is possible by (9.2) and continuity of measure. Applying the portmanteau inequality to $U_\eta = \{u < \eta,\ v < \eta\}$ (for each fixed $\eta > 0$, $\pi(U_\eta) \geq z$) and to $V = \{uv > \eta_0\}$ ($\pi(V) = w$), and choosing $\eta_n \downarrow 0$ slowly by a diagonal argument, we get for all large n:

$$\#J_n^0 \geq \tfrac{z}{2}n, \quad J_n^0 = \{j : \kappa_j < \eta_n,\ \xi_j < \eta_n\}; \qquad \#J_n^1 \geq \tfrac{w}{2}n, \quad J_n^1 = \{j : \kappa_j\xi_j > \eta_0\}.$$

For large n also $\eta_n^2 < \eta_0$, so $J_n^0 \cap J_n^1 = \emptyset$. Let $c_0 = \min(z, w)/3$ and $m_n = \lfloor c_0 n \rfloor$, and choose distinct $p_1, \ldots, p_{m_n} \in J_n^1$ and $q_1, \ldots, q_{m_n} \in J_n^0$. (For the finitely many small n where this is impossible, take $m_n = 0$.)

Step 2: the sequences. Let $e_n = 1/n$ and fix $t \geq 0$. Define A_n and B_n as follows. On each pair of coordinates (p_i, q_i) put the 2×2 blocks $\kappa_{p_i} A_{e_n}$ and $\xi_{p_i} B_{e_n,t}$ from (5.1). On every unpaired coordinate j put the diagonal entries $\tilde{\kappa}_j = \max(\kappa_j, e_n^2)$ and $\tilde{\xi}_j = \max(\xi_j, e_n^2)$. All other entries are zero. Since $\kappa_{p_i}, \xi_{p_i} > 0$, both matrices are HPD. Compared with $D_n(\kappa)$ and $D_n(\xi)$, the changes form, after a permutation, a direct sum of 1×1 and 2×2 blocks, so

$$\|A_n - D_n(\kappa)\| \leq Ke_n^2 + \eta_n, \qquad \|B_n - D_n(\xi)\| \leq Kte_n + K(1 + t^2)e_n^2 + \eta_n + e_n^2,$$

both tending to 0. By (G2) and (G3), $\{A_n\} \sim_{\mathrm{GLT}} \kappa$ and $\{B_n\} \sim_{\mathrm{GLT}} \xi$, and the norms are uniformly bounded.

Step 3: the trace. By (2.2) and (5.2) the mean is the direct sum of the scalars $\sqrt{\tilde{\kappa}_j \tilde{\xi}_j}$ and of the blocks $s_{p_i}(A_{e_n} \# B_{e_n,t})$, whose trace is $s_{p_i} c(t)(1 + (1 + t^2/2)e_n^2)$. Using $\sqrt{a} - \sqrt{b} \leq \sqrt{a - b}$ for $a \geq b \geq 0$ and $\tilde{\kappa}_j\tilde{\xi}_j - \kappa_j\xi_j \leq e_n^2(\kappa_j + \xi_j) + e_n^4 \leq e_n^2(K + 1)$, we get

$$\frac{\mathrm{tr}(A_n \# B_n)}{n} - \frac{1}{n}\sum_{j=1}^{n} s_j \leq e_n\sqrt{K+1} + \frac{1}{n}\sum_{i=1}^{m_n} s_{p_i}\Big[c(t)(1 + (1 + \tfrac{t^2}{2})e_n^2) - 1\Big], \tag{9.3}$$

where the terms $s_{q_i} \geq 0$ of the Riemann sum were dropped. Take $t = 3/2$, so $c(t) = 4/5$. For large n the bracket is at most $-1/10$, and $s_{p_i} > \sqrt{\eta_0}$, so

$$\limsup_n \frac{\mathrm{tr}(A_n \# B_n)}{n} \leq \int_0^1 \sqrt{\kappa\xi}\, dx - \frac{c_0\sqrt{\eta_0}}{10} < \int_0^1 \sqrt{\kappa\xi}\, dx,$$

because $\sqrt{\kappa\xi}$ is Riemann integrable. If $\{A_n \# B_n\} \sim_{\mathrm{GLT}} \sqrt{\kappa\xi}$, then by (G1), the uniform bound $\|A_n \# B_n\| \leq (\|A_n\| + \|B_n\|)/2$, and a continuous compactly supported test function equal to x on the spectra, the normalised traces would converge to $\int_0^1 \sqrt{\kappa\xi}$. This proves (i).

(ii) Use $t = 0$ for even n and $t = 3/2$ for odd n. For $t = 0$, $B_{e,0} = A_e$ and the block mean is $s_{p_i} A_{e_n}$, so the left side of (9.3) lies between $-\frac{1}{n}\sum_i s_{q_i} \geq -\eta_n$ and $e_n\sqrt{K+1} + Ke_n^2$; the normalised traces tend to $\int \sqrt{\kappa\xi}$ along even n, but stay below it by a fixed amount along odd n. Since the means are uniformly bounded, they have no eigenvalue distribution. □

Corollary 10 (dichotomy). Let $\kappa, \xi \geq 0$ be bounded, Riemann-integrable functions on $[0, 1]^d$, regarded as scalar GLT symbols independent of θ. Then the following are equivalent:

1. for every pair of HPD sequences $\{A_n\} \sim_{\mathrm{GLT}} \kappa$, $\{B_n\} \sim_{\mathrm{GLT}} \xi$, one has $\{A_n \# B_n\} \sim_{\mathrm{GLT}} \sqrt{\kappa\xi}$;
2. $|\{\kappa + \xi = 0\}| = 0$ or $|\{\kappa\xi > 0\}| = 0$.

Proof. If $|\{\kappa + \xi = 0\}| = 0$, Theorem 5 applies; if $|\{\kappa\xi > 0\}| = 0$, Corollary 8 applies. Otherwise Theorem 9 gives a counterexample. □

For arbitrary scalar or block GLT symbols, depending on (x, θ), the formula holds if $\kappa + \xi \succ 0$ almost everywhere (Theorem 5) or $\kappa\#\xi = 0$ almost everywhere (Corollary 8). For bounded Riemann-integrable scalar symbols depending only on x, these conditions are also necessary for the formula to hold for every HPD realisation (Corollary 10). For scalar symbols the two conditions mean "no common zero set of positive measure" and "essentially disjoint supports". For block symbols they mean $\ker\kappa \cap \ker\xi = \{0\}$ and $\operatorname{ran}\kappa \cap \operatorname{ran}\xi = \{0\}$ almost everywhere; neither is implied by the absence of a common zero set (for example $\kappa = \xi = \operatorname{diag}(1, 0)$ everywhere, as in Remark 6.2). Necessity for symbols depending on θ, and for block symbols, is open.

When is the symbol formula safe?

- *Safe:* $\kappa + \xi \succ 0$ almost everywhere (Theorem 5). For scalar symbols this says the two symbols have no common zero set of positive measure; isolated zeros, such as $\theta = 0$ for a Laplacian, are harmless. For block symbols it says $\ker\kappa \cap \ker\xi = \{0\}$ a.e. The same holds for every Kubo–Ando mean (Corollary 6).
- *Safe:* $\kappa\#\xi = 0$ almost everywhere (Corollary 8): for scalar symbols, essentially disjoint supports; for block symbols, trivial intersection of the ranges a.e. The means are then zero-distributed.
- *Always true:* the eigenvalues of the means never exceed what the regularised symbol $\kappa\#\xi$ predicts, in the sense of stochastic domination (Theorem 7).
- *Unsafe (proved for bounded Riemann-integrable scalar symbols of x; open in general):* a common zero region of positive measure together with a common support of positive measure (Theorem 9). Typical sources are degenerate diffusion with coefficients vanishing on a subdomain, masked or windowed operators, and covariance models that share a null space. Couplings of size $O(1/N)$ between range and kernel, comparable to discretisation error, then decide the output. For inputs approaching projections, Theorem 4 and Appendix A show that the output spectrum can be anywhere in the envelope (A.1).
- In the unsafe case no numerical test on the symbols can certify the formula: the counterexample inputs are indistinguishable from "good" ones at the symbol level.

10. Conclusion and open problems

The boundary behaviour of the matrix geometric mean is governed by one rank inequality (Theorem 1). Limits of means sit below the regularised mean and can be anything in the rank-constrained part of Ando's feasible set. In geometric terms: the midpoint of the affine-invariant geodesic between two positive definite matrices is not determined by the endpoints once these reach the boundary of the cone, which is one reason why fixed-rank geometries must be built separately [25] (we claim no equivalence with those constructions). For GLT sequences with arbitrary scalar or block symbols, the symbol formula holds if $\kappa + \xi \succ 0$ almost everywhere or $\kappa\#\xi = 0$ almost everywhere, and in every case the regularised symbol is an upper envelope (Theorem 7). For bounded Riemann-integrable scalar symbols depending only on x, these conditions are also necessary for the formula to hold for every HPD realisation. Theorem 4 and Appendix A describe the attainable output freedom for inputs approaching projections.

Open problems:

1. *Structured classes.* Does closure hold for scalar sequences in the algebra generated by fixed-symbol Toeplitz matrices $T_n(f)$, diagonal samplings $D_n(a)$ and small terms $g(n)T_n(f)$? Our scalar counterexamples need a non-local coupling (Remark 6.1); heuristically, fixed-symbol operators couple $\{a = 1\}$ and $\{a = 0\}$ only through $o(n)$ effective rank.
2. *Extra data.* Which asymptotic data beyond the two symbols, for example a joint symbol of the coupling between range and common kernel at scale $1/N$, determine the output symbol? Appendix A and Theorem 4 suggest a contraction-valued field.
3. *The general dichotomy.* Extend Corollary 10 to symbols depending on (x, θ) and to block symbols.
4. *Exact output sets.* Characterise all limiting spectral laws of $A_n \# B_n$ for given input symbols, extending Appendix A from projections to general symbols.
5. *Other means.* Rank-type descriptions of the boundary limits of the harmonic mean, the weighted means $\#_t$, other Kubo–Ando means and Karcher means of more than two matrices; a quantitative version of Theorem 1 controlling the distance to the set of limits.

11. Verification, scope and novelty boundary

Proofs. Every result above has a complete written proof. The quantified statements (all matrices, all dimensions, all sequences) rest on those proofs.

Exact checks. The script `verify_exact.py` (Python standard library only; output written to a path given on the command line) performs 1,030 exact checks in rational and Gaussian-rational arithmetic: formula (5.2) for many (e, t); the construction of Section 3.3 on complex, non-commuting data (including empty blocks), mapped through fixed complex congruences, with the support property (3.2), the rank identity (3.3), exactness of every approximating mean via (2.1), and strictly decreasing errors consistent with $O(\delta)$; the role of each condition in Theorem 1; Corollaries 2–3 on complex data, including the lower endpoint; formula (7.4) under complex unitary conjugation, with odd-size padding; the assemblies (6.1)–(6.6) for $N = 2, \dots, 14$; finite instances of a variant of the construction of Theorem 9 (for $\kappa = 4x$, $\xi = x$ on $(0, 1/2]$, both zero on $(1/2, 1]$, with $\eta_0 = 1/16$ and $\lfloor 3n/16 \rfloor$ pairs taken from the smallest admissible indices instead of the proof's $\lfloor c_0 n \rfloor$, $n = 16, \dots, 256$; the paired blocks are checked exactly and the trace is summed block by block; the normalised traces approach $299/1280 < 1/4 = \int \sqrt{\kappa\xi}$); and the block example of Remark 6.2. Each check raises an error on failure (no `assert` statements, so `python -O` does not disable them), and the option `--self-test-mutation` corrupts formula (5.2) to show that the failure is detected. These checks test identities and finite assemblies; they do not test limits or quantifiers.

Sources. The arXiv version of [17], the thesis [18], the article [16] and the two ETNA papers [14], [15] were accessible. For the published version of [17] the relevant locations and the short passages quoted above were verified; in addition, a separate AI-assisted reading of the published full text confirmed that the hypotheses of its Theorem 3 (pp. 8–9) and Conjecture 1 (p. 19) admit our examples and that its proof contains the attacked identification (p. 10). The published statements were not transcribed in full (Section 6.3). The 2017 monograph [12] was not accessible, so the status of its Conjecture 10.1 is stated conditionally. Jones [19] is a self-hosted, book-length preprint dated October 2026; only §4.12 (printed pp. 105–110) was read in detail.

Novelty. A dated, bounded literature search (recorded in `source_novelty_ledger.md` and `novelty_report.md`) found no earlier statement of the rank criterion (2.4), of the GLT counter-examples, of Theorems 4, 7 and 9, or of Appendix A. This does not establish priority. The discontinuity mechanism, the maximal characterisation, the envelope $X \preceq A \# B$, the dilation description

of the set of limits and the sum normalisation are earlier work and are credited above. A public AI-generated review on Pith, dated 15 August 2026 (https://pith.science/paper/2505.03256), had already questioned the a.c.s. step in the arXiv version without giving a counterexample; we claim no priority for noticing a possible gap. We have not corresponded with the authors of [17].

Declarations

Provenance and status. This manuscript and its code were developed with AI assistance (ChatGPT for the original draft; Claude-based agents for the revision). Separate AI agents checked the proofs and the literature. The work has not undergone independent human peer review or formal proof-assistant verification. It is released as an unrefereed candidate.

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Archive and DOI. Version 0.1.0-candidate of this package (manuscript, code and records) is archived at Zenodo, DOI 10.5281/zenodo.23912963 (all versions: DOI 10.5281/zenodo.23912962), and on GitHub at https://github.com/ipitchford/matrix-geometric-mean-glt-closure.

Appendix A. A sharp spectral envelope at a projection

This appendix concerns growing dimensions. Let P_N be orthogonal projections on $\mathbb{C}^N$ of rank r_N, with $r_N/N \to p \in [0,1]$, and $q_N = N - r_N$.

Theorem A.1. Let $A_N, B_N \succ 0$ with $\|A_N - P_N\| + \|B_N - P_N\| \to 0$, and let μ_N be the empirical eigenvalue measure of $X_N = A_N \# B_N$. Every weak subsequential limit of μ_N has the form

$$\mu = (1-p)\delta_0 + (2p-1)_+\delta_1 + \min(p, 1-p)\,\nu, \tag{A.1}$$

with ν a probability measure on $[0,1]$ (irrelevant when its coefficient is 0). Conversely, every measure of this form is the limit of μ_N for suitable such A_N, B_N, for the given projections.

So a null space occupying a fraction $1-p$ of the dimension can alter at most that fraction of the spectrum, and this is attained (Figure A.1). At $p = 1/2$ the whole nonzero half of the spectrum is undetermined; this is Theorem 4.

Proof of necessity. Write $P_N = \mathrm{diag}(I_{r_N}, 0_{q_N})$ by a unitary change of basis. Since $0 \prec X_N \preceq (A_N + B_N)/2 = P_N + o(1)$, the spectra are bounded; the compression of X_N to $\ker P_N$ is $o(1)$, and so is the off-diagonal block, because $\|X_{12}\|^2 \le \|X_{11}\|\,\|X_{22}\|$ for a positive semidefinite block matrix. If both blocks are nonempty write $A_N = \left(\begin{smallmatrix} a_N & b_N \\ b_N^* & d_N \end{smallmatrix}\right)$ and $T_N = \left(\begin{smallmatrix} a_N^{-1/2} & -a_N^{-1} b_N \\ 0 & I \end{smallmatrix}\right)$. Then $\|T_N - I\| \to 0$ and $T_N^* A_N T_N = \mathrm{diag}(I, S_N)$ with $S_N = d_N - b_N^* a_N^{-1} b_N \succ 0$. Put $Y_N = T_N^* X_N T_N$, $Z_N = T_N^* B_N T_N$, and $C_N = (Y_N)_{11}$. Two implicit steps: (i) $C_N \succeq 0$, as a compression of the positive matrix Y_N; (ii) $Y_N - X_N = o(1)$ because $T_N \to I$ and X_N is bounded, so $(Y_N)_{12}$ and $(Y_N)_{22}$ are $o(1)$ and $X_N = \mathrm{diag}(C_N, 0) + o(1)$. By (2.2), $Y_N = \mathrm{diag}(I, S_N) \# Z_N$, and the $(1,1)$ block of (2.1) gives

$$(Z_N)_{11} - C_N^2 = (Y_N)_{12} S_N^{-1} (Y_N)_{12}^* =: R_N \succeq 0, \qquad \mathrm{rank}\, R_N \le q_N.$$

Also $(Z_N)_{11} = I + E_N$ with $\|E_N\| \to 0$, since $Z_N \to P_N$. Thus $C_N^2 = I - R_N + E_N$. The matrix $I - R_N$ has at least $(r_N - q_N)_+$ eigenvalues equal to 1 and all eigenvalues at most 1; by Weyl's inequality, at least $(r_N - q_N)_+$ eigenvalues of C_N^2, hence (by (i)) of C_N, are within $o(1)$ of 1, and all lie in $[0, 1 + o(1)]$. By (ii) and Weyl's inequality, X_N has at least q_N eigenvalues within $o(1)$ of 0 and, disjointly, at least $(r_N - q_N)_+$ within $o(1)$ of 1, and all its eigenvalues lie in $[-o(1), 1 + o(1)]$. If a block is empty the same conclusions are immediate.

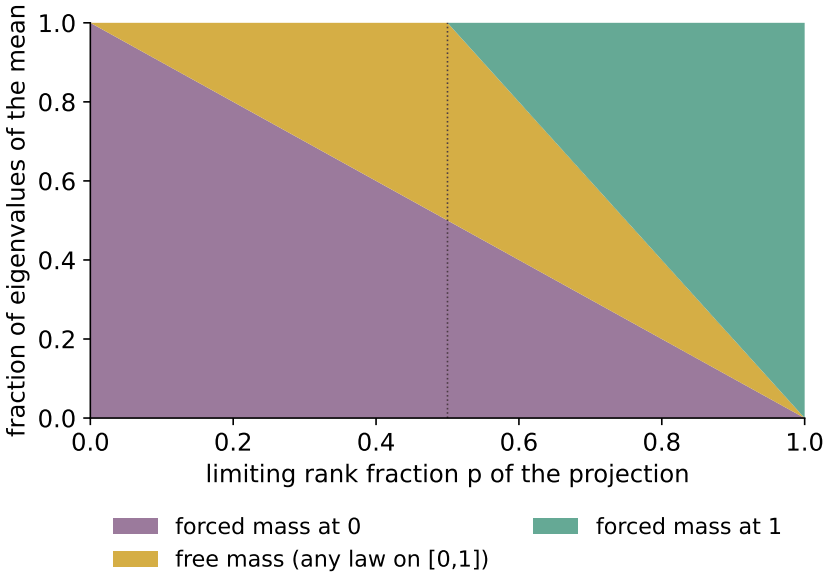


Figure A.1. The envelope (A.1): for each limiting rank fraction p, the share of eigenvalues of the mean forced to 0, forced to 1, and free to have any distribution on $[0, 1]$.

Portmanteau step. Let μ be a subsequential weak limit and $\eta > 0$. For the closed sets $[-\eta, \eta]$, $[1-\eta, 1+\eta]$ and $[-\eta, 1+\eta]$, $\mu(F) \geq \limsup \mu_N(F)$ gives $\mu([-\eta, \eta]) \geq 1-p$, $\mu([1-\eta, 1+\eta]) \geq (2p-1)_+$ and $\mu([-\eta, 1+\eta]) = 1$. Letting $\eta \downarrow 0$, $\mu(\{0\}) \geq 1-p$, $\mu(\{1\}) \geq (2p-1)_+$ and μ is supported in $[0, 1]$. Since $(1-p) + (2p-1)_+ + \min(p, 1-p) = 1$, the measure $\mu - (1-p)\delta_0 - (2p-1)_+\delta_1 \geq 0$ has mass $\min(p, 1-p)$ and is supported in $[0, 1]$, which is (A.1).

Proof of attainability. Let $d_N = \min(r_N, q_N)$, choose numbers $c_{N,1}, \ldots, c_{N,d_N} \in [0, 1]$ whose empirical measures tend to ν (when $d_N/N \to 0$ nothing is needed), and pair d_N range directions of P_N with d_N null directions. Let $\eta_N \downarrow 0$, $\eta_N \leq 1$, and for each pair replace c by $\hat{c} = (c+\eta_N)/(1+\eta_N)$, put $t = 2\sqrt{\hat{c}^{-2} - 1}$ (so $\sqrt{4+t^2} = 2/\hat{c}$ and $t \leq 2(1+\eta_N)/\eta_N$), $\varepsilon_N = \eta_N^2$, and use the pair $(A_{\varepsilon_N}, B_{\varepsilon_N,t})$ of (5.1). Uniformly in $c \in [0, 1]$, $\|B - \mathrm{diag}(1, 0)\| \leq \varepsilon_N t + \varepsilon_N^2(1+t^2) \leq 21\eta_N$, $\|A - \mathrm{diag}(1, 0)\| = \eta_N^4$, and by (5.2) $\|A\#B - \mathrm{diag}(c, 0)\| \leq |\hat{c} - c| + \varepsilon_N t\hat{c}/2 + \varepsilon_N^2(2+t^2)\hat{c}/2 \leq 8\eta_N$. On the remaining range directions take $A = B = 1$, on the remaining null directions $A = B = \varepsilon_N^2$. Direct sums preserve these uniform bounds, and undoing the unitary change of basis completes the proof. □

For the special case of identical density matrices in fixed dimension see Corollary 3.

Appendix B. The dilation view and its relation to Theorem 1

This appendix records a general description without claiming priority for it. Its attainability construction is Jones's Lemma 4.12.3 [19] (printed pp. 106–107), stated there for every Kubo–Ando connection; the converse is an elementary compactness argument.

Fix N and a Kubo–Ando mean σ with function ϕ as in (8.4). Given $A, B \succeq 0$, apply a fixed invertible congruence so that

$$S = A + B = P = I_s \oplus 0_k, \qquad A = H \oplus 0_k, \qquad B = (I_s - H) \oplus 0_k, \qquad 0 \preceq H \preceq I_s.$$

where $k = N - s$ is the dimension of the common kernel. Let $\mathcal{C}_\sigma(A, B)$ be the set of limits of $A_j \sigma B_j$ with $A_j, B_j \succ 0$, $(A_j, B_j) \to (A, B)$.

Proposition B.1. In these coordinates

$$\mathcal{C}_\sigma(A,B) = \{P\phi(T)P : \ 0 \preceq T \preceq I_N, \ PTP = H \oplus 0_k\}. \tag{B.1}$$

Proof. Necessity: with $S_j = A_j + B_j$ and $T_j = S_j^{-1/2} A_j S_j^{-1/2}$, we have $0 \prec T_j \prec I$, so a subsequence converges to some T with $0 \preceq T \preceq I$; since $S_j^{1/2} \to P$, $A = PTP$, and $A_j \sigma B_j = S_j^{1/2} \phi(T_j) S_j^{1/2} \to P\phi(T)P$. Attainability: for $0 < \varepsilon < 1/2$ put $T_\varepsilon = (1-2\varepsilon)T + \varepsilon I$, $S_\varepsilon = P + \varepsilon^2(I-P)$, $A_\varepsilon = S_\varepsilon^{1/2} T_\varepsilon S_\varepsilon^{1/2}$ and $B_\varepsilon = S_\varepsilon^{1/2}(I-T_\varepsilon)S_\varepsilon^{1/2}$. Both are positive definite and tend to A, B, and $A_\varepsilon \sigma B_\varepsilon = S_\varepsilon^{1/2}\phi(T_\varepsilon)S_\varepsilon^{1/2} \to P\phi(T)P$. □

Relation to Theorem 1. For $\sigma = \#$, $\phi(t) = \sqrt{t(1-t)}$. One inclusion between (B.1) and condition 2 of Theorem 1 is immediate through the factorisation $M = ZZ^*$: given T, put $F = PT^{1/2}$ and $G = P(I-T)^{1/2}$. Then $FF^* = PTP = A$, $GG^* = P - PTP = B$, and $FG^* = PT^{1/2}(I-T)^{1/2}P = P\phi(T)P =: X \succeq 0$, so

$$M(A,B,X) = \begin{pmatrix} F \\ G \end{pmatrix} \begin{pmatrix} F \\ G \end{pmatrix}^* \succeq 0, \qquad \operatorname{rank} M(A,B,X) \le N.$$

For the reverse inclusion we do not have a two-line elimination; it follows by combining the two complete characterisations: condition 2 implies, by the construction of Section 3, that X is a limit, and Proposition B.1 then supplies T. Thus Theorem 1 is the closed form of the dilation description for the geometric mean: it replaces the existence of the auxiliary contraction T by block positivity and one rank inequality. Fixed dimension is essential to the compactness argument; limits in growing dimension need separate estimates, as in Appendix A.

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